Building Management System

AI/ML Document

Chiller Temperature Setpoint Reset Control and Cooling Tower Leaving Temperature Setpoint Reset

Project	CSIT SP2, Science Park Road
Referenced by	Chiller System Control Description, Rev 5, 1 April 2026, sections 3.11 and 4.7
Document number	To be assigned by the BMS contractor
Submitted by	RoyceMedia Technologies Pte Ltd
Revision	1 · 16 October 2026 · initial proposal draft
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Revision history

Rev	Date	Details
1	16 Oct 2026	Initial proposal draft

1. What this document is

Section 3.11 of the Chiller System Control Description says: “Refer to AI/ML Document.” This is that document.

It explains how the chilled water supply temperature setpoint will be reset with the help of an AI/ML system: what we propose, what data the system reads, how it decides the setpoint, the limits it keeps to, what happens if it stops, and how the saving is measured. The same system also recommends the cooling tower leaving water setpoint, which the Control Description resets by formula in section 4.7, so that value is covered here in the same way. Everything else in the Control Description stays as it is.

2. The plant today

The plant serves an existing building at Science Park Road. It was built with variable-speed drives on every chiller, pump and tower fan, a Honeywell BMS, and one-minute logging of every point, which is what makes an advisory system practical without any new hardware.

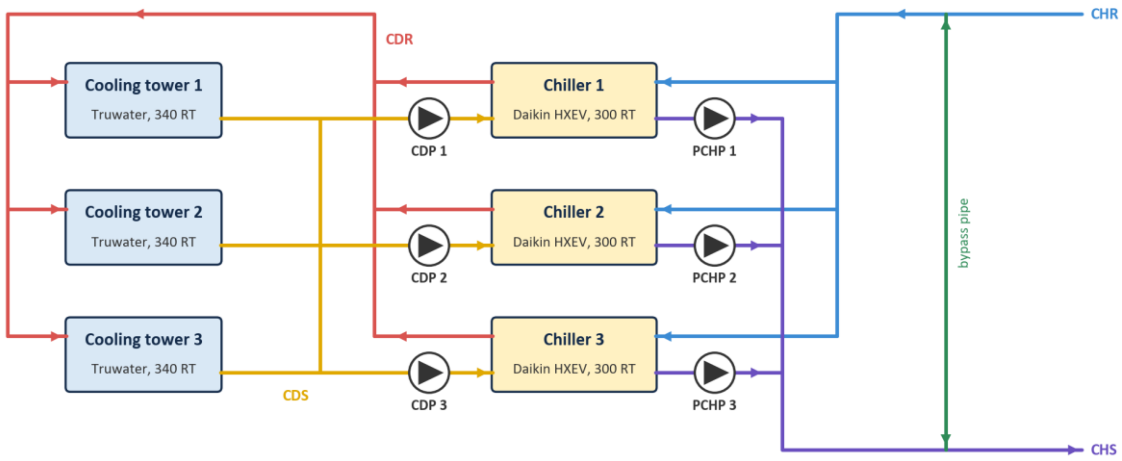


Figure 1. The plant as installed.

Three cooling towers, three condenser water pumps, three chillers and three chilled water pumps. CDP are the condenser water pumps and PCHP the chilled water pumps. CDS and CDR are the condenser water supply and return, CHS and CHR the chilled water supply and return.

Equipment	Count	Make and model	Rating	Speed control
Chillers	3	Daikin HXEV490TSTGG, water-cooled magnetic-bearing centrifugal	300 RT each	Variable-speed drive, built in
Chilled water pumps	3	Paragon PA-100-32	36 L/s each	Variable-speed drive
Condenser water pumps	3	Paragon PA-125-26	52 L/s each	Variable-speed drive
Cooling towers	3	Truwater iVXS-2220A-3LIA	340 RT each	EC fan, 40 to 100 %

3. What we propose

Today the chilled water supply setpoint is fixed at 10 °C from morning to night. The cooling coils need water that cold only at the peak of the day. The rest of the time they have capacity to spare and would be satisfied with slightly warmer water. Warmer chilled water is less work for the chillers: about 2 to 3 % less chiller power for each degree the setpoint is raised.

The signal that a coil has capacity to spare is its chilled water valve. A valve at 60 % open has margin; a valve at 100 % is already asking for everything the coil can give. So the system reads the air side as well as the plant, and raises the setpoint only while every monitored coil still has margin and every AHU still holds its supply air temperature.

The cooling towers are the second opportunity. Today their leaving water setpoint comes from a BMS formula, 0.5 × wet-bulb + 27, held between 23 and 29.5 °C. For Singapore wet-bulbs the formula gives more than 29.5 °C, so the setpoint sits at the ceiling all day. Colder condenser water is less work for the chillers, about 3 % less chiller power for each degree, but the fans use more power to make it. For most of the day a lower setpoint saves more on the chillers than it costs on the fans.

We propose an advisory AI/ML system that:

- reads the plant, the cooling towers and the AHU valve positions from the BMS, every minute, read-only;
- every 15 minutes works out two values: the highest chilled water setpoint, between 10.0 and 11.5 °C, that still keeps every cooling coil satisfied, and the cooling tower leaving water setpoint, between wet-bulb + 2 °C and 29.5 °C, that gives the lowest chiller plus fan power;
- shows each value to the plant operator with the expected saving and a one-line reason;
- the operator enters the values at the BMS head-end. The AI/ML system does not write to the BMS and does not control any equipment.

If the system stops, if its data is in doubt, or if the operator disagrees, the setpoints stay at today's values, or return to them, and the plant runs exactly as it does today.

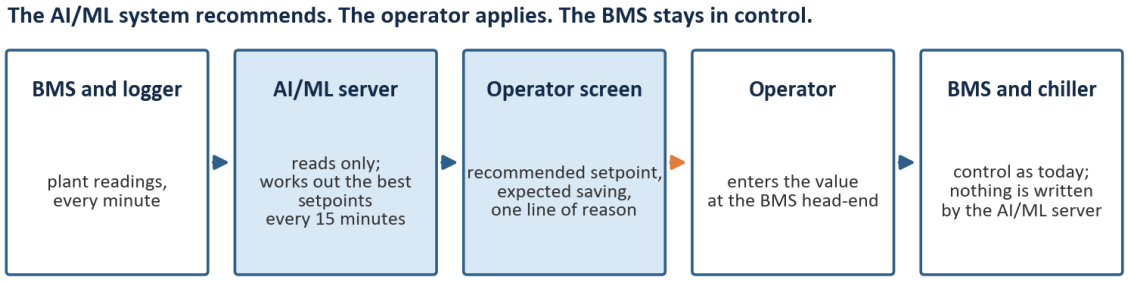


Figure 2. How the advisory reset works.

What to expect

Recommended value	Today	Recommended range	Expected effect
Chilled water supply setpoint	Fixed at 10.0 °C	10.0 to 11.5 °C, moving by up to 0.3 °C every 15 minutes	Chiller power down about 2 to 3 % for each degree raised; chilled water pumps up a little
Cooling tower leaving water setpoint	0.5 × wet-bulb + 27, held between 23 and 29.5 °C; in practice 29.5 °C	Wet-bulb + 2 °C up to 29.5 °C, never below 23 °C, moving by up to 0.5 °C every 15 minutes	Chiller power down about 3 % for each degree lowered; tower fans up

Two to eight recommendations a day is normal. Each one is a single number for the operator to key in. The percentages above are from published studies of plants like this one; the plant's own figures come from its data once the models are fitted.

Checked before every recommendation

A setpoint change in the plant room is felt on the air side and inside the chillers. Every recommendation is checked against both before it is shown, and the operator screen shows each check with its live value, so the operator sees the whole site and not just the plant.

Recommended value	What is checked, every 15 minutes	If a check fails
Chilled water supply setpoint	• Air side: every monitored AHU chilled water valve below 90 % open • Air side: every AHU holding its supply air temperature, within 0.5 °C of its setpoint • Air side: zone temperature and humidity within their limits, where the BMS has them • Air side: DOAS units holding their supply dew point, where available • Plant: chillers not at full load, the header reaching its setpoint, pumps not at maximum speed • Next hour: the forecast load still within capacity at the higher setpoint	• No rise is recommended • If a valve is above 95 % or an AHU is not holding its supply air temperature, the setpoint comes down 0.3 °C per cycle • If the air-side points are not available or are stale, the setpoint stays at 10.0 °C
Cooling tower leaving water setpoint	• Chillers: condenser water within the chiller's range from the submittal • Chillers: condenser approach, load and any high-pressure or surge warning, read each cycle • Towers: the setpoint at least 2 °C above the wet-bulb for the next hour, fans not already at 100 % • Limits: never above 29.5 °C, the BMS maximum and today's value, never below 23 °C	• No change is recommended • If a chiller is near a limit, the tower setpoint is held • If the weather station or the tower points are stale, the setpoint goes back to the BMS formula value

4. How the chilled water setpoint is chosen

The rule follows the cooling-coil valves, not the outdoor temperature. This is the method ASHRAE recommends for variable-flow plants, because a coil whose valve is not fully open still has cooling and dehumidification capacity in hand.

- If every monitored AHU valve has been below 90 % open for 15 minutes, the setpoint may be raised by 0.3 °C.
- If any monitored valve is above 95 % open, the setpoint is lowered by 0.3 °C.
- Between 90 % and 95 %, the setpoint is held.
- The setpoint never goes above 11.5 °C and never below the design value of 10.0 °C.

Within that rule, the AI/ML model picks the value that gives the lowest chiller plus pump power, using the chiller and pump curves explained below. The reset is active during staffed hours, to be confirmed with the owner. Outside those hours the recommended setpoint is 10.0 °C.

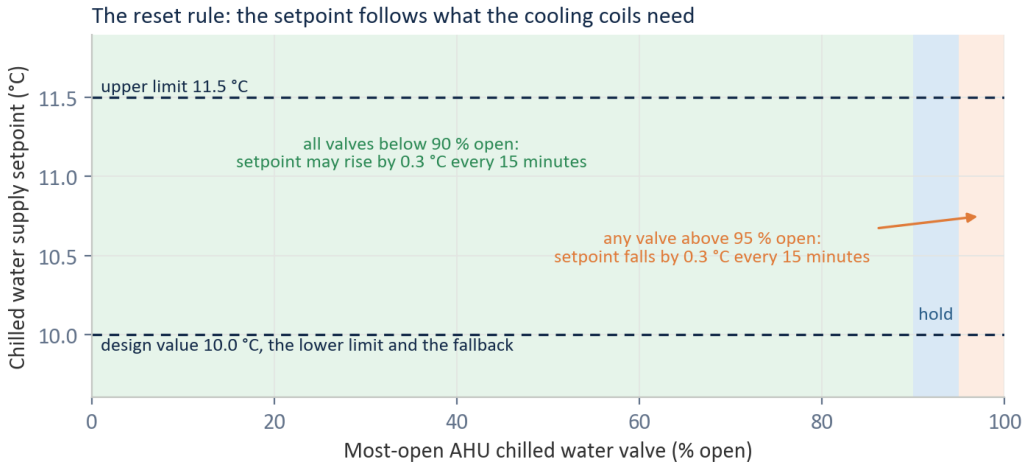


Figure 3. The reset rule.

Values that can be adjusted

Value	Initial setting	Who adjusts
Lower limit, also the fallback value	10.0 °C	Fixed by design
Upper limit	11.5 °C	Owner
Step per 15 minutes	0.3 °C	OptimusAI, at go-live
Valve position to allow a rise / to force a fall	90 % / 95 % open	OptimusAI, at go-live
Hold time before a rise	15 minutes	OptimusAI, at go-live
AHU supply air temperature tolerance	0.5 °C above its setpoint	OptimusAI, at go-live
Staffed hours	TBC	Owner

5. How the cooling tower setpoint is chosen

Lowering the tower setpoint makes the condenser water colder. Each chiller then has less lift to work against and uses less power. The fans have to run faster to make the colder water, and fan power rises steeply as the water gets close to the wet-bulb temperature. Somewhere between the two is the setpoint with the lowest total. That point moves with the wet-bulb and the load, which is why a fixed formula cannot find it and the AI/ML system works it out every 15 minutes.

- Every 15 minutes the system tries each tower setpoint in 0.5 °C steps between wet-bulb + 2 °C and 29.5 °C and keeps the one with the lowest chiller plus fan power.
- The setpoint never goes below 23 °C, the BMS minimum, and never above 29.5 °C, the BMS maximum, which is today's value. The towers are only ever asked for colder water than today, never warmer.
- Each cycle the chillers' condenser approach, load and warnings are read. If a chiller is near a limit, the tower setpoint is held.
- It moves by at most 0.5 °C per 15 minutes, so the chillers and towers settle between steps.
- The number of tower cells running stays as the BMS stage table sets it. The fans follow the setpoint as they do today.

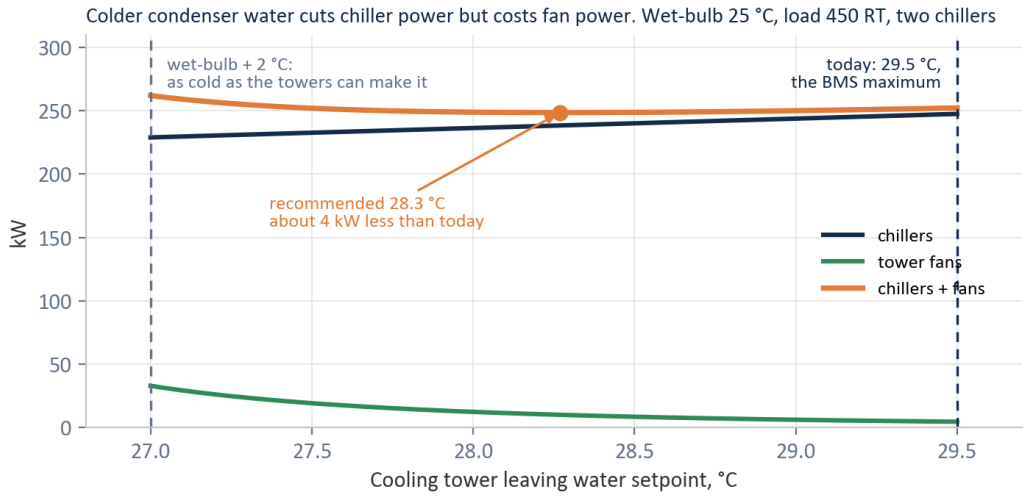


Figure 4. The trade-off behind the tower setpoint, illustrative.

Values that can be adjusted

Value	Initial setting	Who adjusts
Closest approach to wet-bulb	2 °C	OptimusAI, at go-live
Chiller condenser water limits	From the chiller submittal	Fixed by the chiller
Lower limit	23 °C, the BMS minimum	Fixed by the BMS
Upper limit, also the fallback value	29.5 °C, the BMS maximum	Fixed by the BMS
Step per 15 minutes	0.5 °C	OptimusAI, at go-live
Staffed hours	TBC	Owner

6. What the system reads

All of the points below already exist on the BMS or the logger. They are read once a minute. Nothing is written back. The BMS object names are listed in the separate Point List Request.

Where	Points	Used for
Plant headers	• Chilled water supply and return temperature • Chilled water flow • Condenser water supply and return temperature • Condenser water flow • Cooling load and plant power as the BMS calculates them	Cooling load and the check that the sensors agree with each other
Each chiller	• Run status and any fault, warning or trip • Power in kW and load in % • Chilled water and condenser water temperatures in and out • Condenser pressure or approach • The setpoint the chiller is currently holding	The chiller performance curve and the chiller-side checks
Chilled water pumps	• Run status, speed and power of each pump	The pump curve
Cooling towers	• Run status, fan speed and power of each cell • Leaving water temperature and the setpoint the towers currently hold	The cooling tower model and the fan curve
Air side	• Valve position of each AHU, or the most-open valve • AHU supply air temperature and its setpoint • Zone temperature and humidity, where the BMS has them • DOAS supply dew point, where available	The chilled water reset rule and the air-side checks
Weather station	• Outdoor temperature and humidity at the cooling towers, from which the wet-bulb is calculated	The tower setpoint range, and the weather and cooling demand forecasts
Plant control	• Stage number • Plant mode • Any point in manual or override	Knowing when a recommendation should not be made

7. The AI optimiser

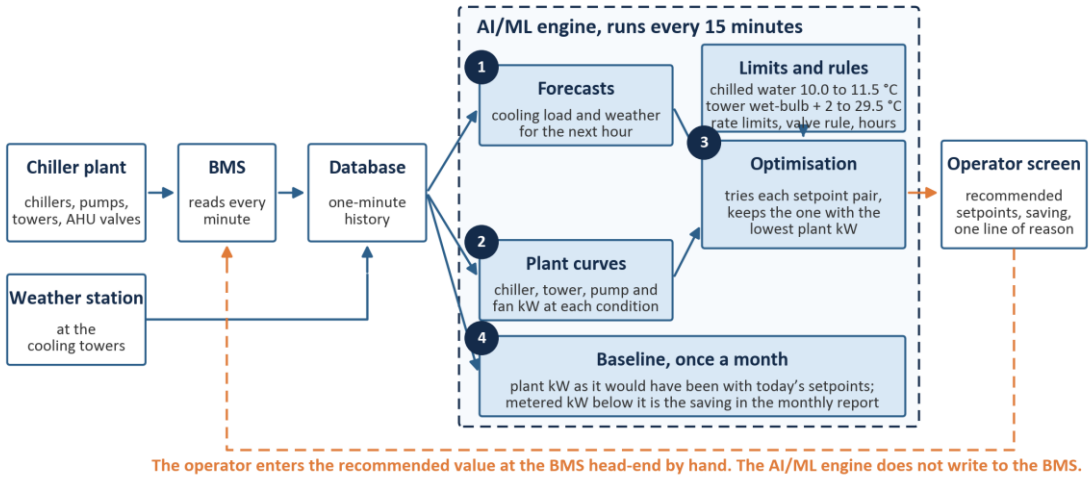


Figure 5. The AI optimiser: how the models work together.

The system builds six models from the plant's own one-minute data. The first fit uses the data logged so far, and the models are refitted every month as more is logged. The chiller, tower, pump and fan curves work out the kW each candidate setpoint saves. The demand and weather forecasts check the hour ahead before a change is recommended. The baseline is used once a month to report the saving.

The most recent two weeks of data are kept aside as test days and are not used for fitting. Each model's accuracy is measured on those days and reported at every refit.

Model	Inputs	Output	Method	Accuracy on test days
Cooling demand forecast	Load history, time of day, day of week, outdoor temperature and humidity	Cooling load for the next hour, in 15-minute steps	Machine learning, gradient-boosted trees	Within 5 %
Weather forecast at the cooling towers	The weather station's recent readings and time of day	Outdoor temperature and humidity for the next hour	Machine learning, gradient-boosted trees, from the station's own readings	Within 1 °C
Chiller performance curve, one per chiller	Chiller load, chilled water supply temperature, condenser water entering temperature	Chiller kW	Physics-informed curve, fitted from the site data	Within 3 %
Cooling tower model, one per cell	Condenser water flow and return temperature, fan speed, wet-bulb	Tower leaving water temperature	Physics-informed effectiveness model, fitted from the site data	Within 0.3 °C
Pump and fan curves	Chilled water flow and pump speed; tower fan speed	Pump kW and fan kW	Physics-informed curves, fitted from the site data	Within 3 %

Model	Inputs	Output	Method	Accuracy on test days
Baseline	Cooling load, outdoor temperature and humidity, time of day, day of week	Plant kW as it would have been with today's setpoints	Machine learning, gradient-boosted trees, fitted on data from before the recommendations start	Within 5 %

How the forecasts are used

Both forecasts look one hour ahead so that a recommendation still makes sense by the time the operator has entered it and the plant has settled.

- **Cooling demand forecast.** Before a higher chilled water setpoint is recommended, the system checks the load coming in the next hour. If the load is about to rise, for example at the morning start-up or the lunch peak, it holds the setpoint rather than raise it and reverse it 15 minutes later.
- **Weather forecast.** The next-hour outdoor temperature and humidity are an input to the cooling demand forecast, and they give the next-hour wet-bulb. The tower setpoint is chosen so that it stays at least 2 °C above the wet-bulb for the whole hour, not just at this minute. If the wet-bulb is 25 °C now and rising to 26.5 °C by noon, the system recommends 28.5 °C, not 27 °C, so the towers do not chase a value they cannot reach.
- **Where the weather forecast comes from.** The forecast is made from the weather station's own last few hours of readings and the time of day; no outside weather feed is used. Singapore weather follows a daily pattern closely enough for a one-hour forecast. An afternoon storm is the exception; when the forecast misses by more than 1 °C the system uses the current readings instead until the forecast recovers.

The chiller performance curve

How much power the reset saves depends on how each chiller's power changes with its load and with the water temperatures on both sides. That relationship is the chiller performance curve, and it is the model behind every recommendation. The left chart shows the shape of the curve; the right chart shows what the reset does to one chiller at a typical load.

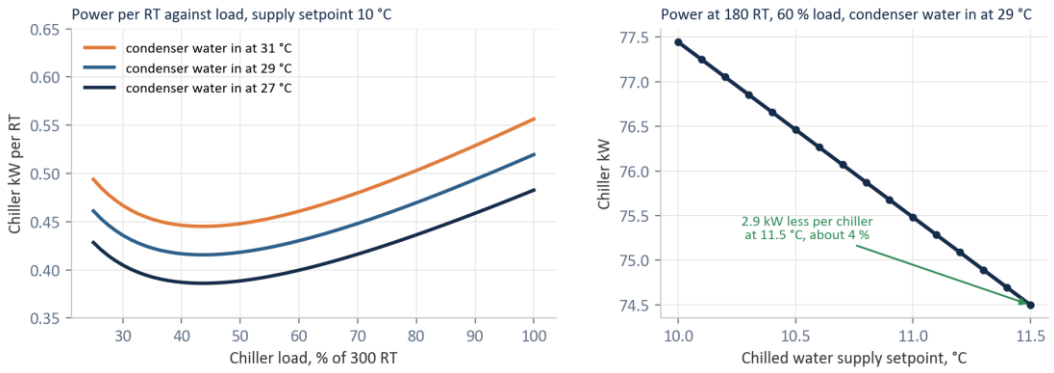


Figure 6. The chiller performance curve, illustrative shape from the catalogue rating.

We fit the curve for each of the three chillers from the plant's own one-minute data: chiller kW against cooling load, chilled water leaving temperature and condenser entering temperature. The fitted curves replace the chart above in the first monthly report. Each curve is checked in three ways before it is used:

- **Against the catalogue.** Daikin's catalogue for the HXE range, the family the site's HXEV chillers belong to, rates the 300 RT model at 0.55 kW per RT at full load at the catalogue's standard rating conditions: chilled water leaving at 6.7 °C and condenser water entering at 29.4 °C. At part load the figure is much lower. The fitted curve must reproduce the catalogue points within a few percent; the site submittal gives the exact figures for the installed model.
- **On the test days.** The most recent two weeks of data are not used for fitting. On those days the curve must predict chiller kW within 3 %.
- **Against the plant meter.** Replaying all three chillers over those days must match the metered energy within 2 %.

The baseline and the saving report

The baseline is a model of plant kW with today's setpoints: chilled water at 10 °C and the tower formula. It is fitted on data logged before the recommendations start, with the cooling load, the weather and the calendar as inputs. Each month it is run on that month's load and weather to give the kW the plant would have used with today's setpoints. The metered kW is subtracted from it, and the difference is the saving reported for the month. The baseline is refitted only if the plant itself changes, for example a chiller replacement, so that every month is measured against the same yardstick.

8. Limits and safeguards

Situation	What the system does
Any monitored AHU valve above 95 % open	Lowers the recommended chilled water setpoint by 0.3 °C per cycle
Any AHU not holding its supply air temperature, or a zone outside its limits	Lowers the recommended chilled water setpoint by 0.3 °C per cycle
A chiller near its condenser limit: high approach, high load or a warning	Tower setpoint held until the chiller has settled
AHU valve data missing or older than 5 minutes	Recommended chilled water setpoint goes to 10.0 °C
Tower setpoint would be below wet-bulb + 2 °C or below 23 °C	Not recommended; the towers cannot make water that cold
Tower cell fault, or a fan in manual	Recommended tower setpoint goes back to the BMS formula value
Chiller fault, trip or lockout	Both recommended setpoints go back to today's values
Sensors disagree: plant heat balance outside ±5 %	No recommendation until the readings agree
Setpoint in manual or override at the BMS	No recommendation for that setpoint

Situation	What the system does
Operator dismisses a recommendation	Logged with the reason; not repeated until conditions change
Outside staffed hours	Recommended setpoints are today's values

Both changes reduce the chiller's lift, so they move the chiller away from its surge limit, not towards it. Daikin's catalogue gives the chiller's working range as 3 to 18 °C for chilled water leaving and 18 to 45 °C for condenser water. The ranges recommended here, 10.0 to 11.5 °C and 23 to 29.5 °C, sit inside them.

9. If the system stops

The AI/ML server has no connection that can change the plant. If the server stops, the network drops, or the screen is switched off, nothing happens to the plant: the BMS keeps the last setpoints the operator entered and runs the sequences in the Control Description. The operator can return the chilled water setpoint to 10.0 °C and the tower setpoint to the BMS formula at any time.

10. What the operator sees

One screen on the plant network. For each of the two values it shows the current setpoint, the recommended setpoint, the expected plant kW before and after, and one line of reason. Beside them are the checks with their live values: the most-open AHU valve, the AHU supply air temperatures, zone humidity where available, the wet-bulb, the chillers' condenser approach and load, and any warning. Two buttons: "Mark applied" after the operator has entered the value at the BMS, and "Dismiss" with a reason. When no change is worth making, the screen says "no change". We expect two to eight operator actions a day.

11. Who does what

OptimusAI	BMS contractor	Owner
• Installs the AI/ML server on the plant network • Fits and checks the models, and refits them every month • Provides the operator screen, training and the monthly saving report	• Confirms the BMS point names and that both setpoints can be entered at the head-end • Provides the logged data	• Confirms the limits and the staffed hours • Nominates the operators who enter the setpoints

12. Assumptions

The advice is only as good as what goes into it. These are the things the system takes as given.

Category	Assumption	What it means for the recommendations
Sensor accuracy	Readings from the BMS and the logger are used as they are, without calibration against reference instruments.	A sensor that reads wrong puts the models and the recommendations wrong by the same amount. The heat balance check catches readings that disagree with each other by more than 5 %, not a sensor that is consistently off.
Weather station	The station at the cooling towers is the only weather source. No outside weather feed is used.	If it reads wrong or fails, the wet-bulb and the forecasts are wrong with it. Tower recommendations stop while its reading is missing or out of range.
Air-side points	AHU valve positions and supply air temperatures, and zone temperature and humidity where they exist, are on the same BMS. To be confirmed by the BMS contractor.	Without them the chilled water setpoint stays at 10 °C and only the tower setpoint is recommended. A valve reading 100 % holds the setpoint at 10 °C, the safe side; a valve reading lower than it really is could allow a rise the coil does not want.
Head-end entry	Both setpoints can be entered at the BMS head-end by the operator. To be confirmed by the BMS contractor.	A setpoint that cannot be entered is not recommended.
Logged data	The models are fitted from the data logged so far and refitted every month.	If the plant has so far run in one way only, the first months are less accurate. Accuracy improves with each refit and is reported each time.
BMS sequences	The sequences, limits and alarms in the Control Description, Rev 5, stay as written.	If the BMS contractor changes any of them, the limits in this document are updated with them.
Chiller limits	Daikin’s catalogue figures and working ranges stand in until the site submittal for the installed model is provided.	The submittal’s condenser water limits replace the catalogue values as the hard limits.
Operator	Nothing changes until the operator enters the value at the BMS.	A recommendation entered late saves less, and one that is dismissed saves nothing. The monthly report counts what was entered.
Plant unchanged	The equipment, the operating hours and the building use stay as they are.	A chiller replacement, a new tenant or new hours changes the baseline, and it is refitted from that point.